TEXORA · LECTURE SHEET

Yarn Geometry

The yarn idealised as a cylinder of partly-packed fibres wound into helices: how diameter follows from count, density and packing; how the twist helix angle follows from diameter and turns per metre; and the twist factor that lets counts be compared.

FM-4203 · FABRIC MECHANICS · TOPIC 2 · FABRIC MANUFACTURING SPECIALIZATION

1. The idealised yarn

Yarn geometry begins with a deliberate simplification, the **idealised yarn**: circular in cross-section, of uniform diameter along its length, with fibres distributed uniformly across the section and each fibre following a perfect helix at constant radius. Real yarns depart from all of this — the section is nearer elliptical once woven, diameter varies, helices migrate — but the idealisation converts count, density and twist into diameters and angles that agree with measurement well enough to design fabrics with (Topic 3) and it exposes the structure of every formula used later.

2. Yarn diameter from count

The yarn envelope contains fibre and air. If the packing fraction is φ and the fibre density ρ, the yarn's effective density is φρ, and the diameter follows from the same area conversion as Topic 1, with ρ replaced by φρ:

$$A = T \times 10^{-3} \div (\varphi\rho)\ \text{mm}^2, \quad d = \sqrt{(4A \div \pi)}$$

WORKED EXAMPLE — DIAMETERS OF TWO COTTON YARNS

Cotton, ρ = 1.52 g/cm³, ring-spun packing fraction φ = 0.60, so φρ = 0.912 g/cm³.

24 tex: $A = 24 \times 10^{-3} \div 0.912 = 0.0263$ mm²; $d = \sqrt{(4 \times 0.0263 \div \pi)}$ = **0.183 mm**.

20 tex: $A = 20 \times 10^{-3} \div 0.912 = 0.0219$ mm²; $d = \sqrt{(4 \times 0.0219 \div \pi)}$ = **0.167 mm**.

Diameter grows only with the *square root* of count: quadrupling the tex only doubles the diameter. These two diameters are the warp and weft of the fabric worked throughout Topic 3 — keep them.

Two cautions belong with the formula. First, φ is not a constant of nature: it rises with twist (twist compresses the yarn), so a hard-twisted 24 tex yarn is slimmer than 0.183 mm and a soft one bulkier. Second, the diameter that matters in fabric geometry is the diameter *in the fabric*, under the flattening

of interlacement — Peirce's model (Topic 3) begins by assuming it unchanged, then corrects for flattening in its refinements.

3. The twist helix

Unwrap one turn of the surface helix onto a plane and it becomes the hypotenuse of a right triangle: the base is the yarn circumference πd, the height is the axial length of one turn, 1/T where T is the twist in turns per unit length. The **surface twist angle** θ — the angle between the surface fibre and the yarn axis — satisfies:

$$\tan\theta = \pi\, d\, T$$

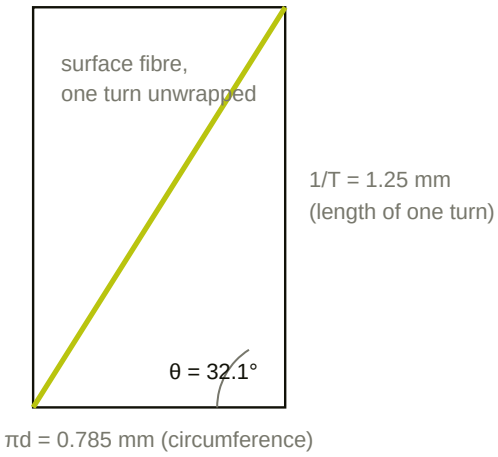


Figure 1 — Development of the surface helix for the worked example: d = 0.25 mm, T = 800 turns/m. One turn unwraps to a triangle of base πd = 0.785 mm and height 1/T = 1.25 mm; tan θ = 0.785 ÷ 1.25 = 0.628, θ = 32.1°. Dimensions drawn to scale.

WORKED EXAMPLE — SURFACE TWIST ANGLE

Yarn diameter d = 0.25 mm, twist T = 800 turns/m = 0.8 turns/mm.
tan θ = π × 0.25 × 0.8 = 0.628 ⇒ θ = **32.1°**.
Consequences that follow immediately: the surface fibre is longer than the yarn it wraps — per unit yarn length it travels 1 ÷ cos θ = 1 ÷ 0.847 = **1.18** times as far — and only cos θ of its direction is along the load. Fibres at the core (radius → 0) have θ → 0: *the twist angle is a radial function*, zero at the axis and maximum at the surface, which is why the surface fibres reach their breaking extension first and yarn failure starts outside and unzips inward.

4. Twist factor: comparing twist across counts

Turns per metre alone cannot compare the "twistiness" of a fine and a coarse yarn — the same T gives a much steeper helix on a fat yarn (tan θ ∝ d). The industry therefore specifies twist relative to count, by

a **twist factor**:

- **English (cotton) system:** twist multiplier α_e = tpi ÷ √Ne.
- **Metric system:** twist factor α_m = turns/m ÷ √tex.

WORKED EXAMPLE — ONE YARN, BOTH SYSTEMS

Ne 30 cotton yarn at twist multiplier α_e = 3.5 (a normal weaving-warp level):

tpi = 3.5 × √30 = 3.5 × 5.48 = **19.2 turns/inch** = 19.2 × 39.37 ≈ **755 turns/m**.

Ne 30 is 590.5 ÷ 30 = 19.7 tex, so the same twist in metric terms: α_m = 755 ÷ √20 (using the nominal 20 tex) = 755 ÷ 4.47 ≈ **169**.

Equal twist factor = equal surface angle (for equal packing), which is the entire point: a mill can carry one α across a count range and keep yarn character — strength utilisation, handle, weaving behaviour — approximately constant while the counts change.

5. What geometry does to strength

Two competing effects of twist are both written in the helix angle. **Cohesion:** the helix presses fibres together (the inward pressure grows with twist), so short fibres grip and the yarn can mobilise fibre strength instead of slipping apart. **Obliquity:** a fibre at angle θ contributes its strength along the yarn axis reduced by a factor that falls roughly as $\cos^2 \theta$, and its breaking extension is consumed partly in rotation. At low twist, cohesion dominates and strength rises; at high twist, obliquity dominates and strength falls. The optimum for cotton ring yarn sits near the angles of Section 3 (θ ≈ 30°), and the same trade-off reappears at fabric level as crimp (Topic 3): *every textile structure buys cohesion with obliquity, and mechanics is the accountancy of that trade*.

Twist liveliness and contraction complete the picture: a twisted yarn stores torsional energy and snarls when slack (the cause of knitted spirality met in TE-3205), and inserting twist shortens the yarn — the surface fibre's 1.18 path ratio of Section 3 is the geometric root of twist contraction in doubling and of the length allowances in production calculations.

Glossary

Idealised yarn — circular, uniform model yarn of perfect helices. **Packing fraction φ** — fibre volume fraction of the yarn envelope. **Surface twist angle θ** — angle between surface fibre and yarn axis, tan θ = πdT. **Twist factor / multiplier** — twist normalised by √count, equalising helix angle across counts. **Obliquity** — loss of axial effectiveness of an inclined fibre. **Twist contraction** — shortening of yarn as twist is inserted.

Key points

• Diameter: $d = \sqrt{(4T \times 10^{-3} \div (\pi\varphi\rho))}$ — 24 tex cotton ≈ 0.183 mm, 20 tex ≈ 0.167 mm at $\varphi = 0.60$; d grows as √tex.
• Helix: $\tan\theta = \pi dT$; worked $\theta = 32.1°$, surface fibre path 1.18× the yarn length; θ grows from 0 at the core to maximum at the surface.
• Twist factor ties twist to count ($\alpha_e = tpi/\sqrt{Ne} = 3.5 \rightarrow 19.2$ tpi ≈ 755 t/m at Ne 30; $\alpha_m \approx 169$) so equal factors give equal angles and similar yarn character.
• Strength versus twist peaks where cohesion gains and obliquity losses balance.

Practice

1. Calculate the diameter of a 30 tex cotton yarn at $\varphi = 0.60$, $\rho = 1.52$ g/cm³.
2. What twist (turns/m) gives a surface angle of 30° on a yarn of diameter 0.20 mm?
3. A yarn of Ne 20 is to be spun at the same twist multiplier as the worked example ($\alpha_e = 3.5$). Find its twist in turns/inch and turns/m, and state why the turns/m differ from Ne 30 at the same multiplier.
4. Explain why a yarn fails from the surface inward, using the radial variation of θ.

Answer key

1. $A = 30 \times 10^{-3} \div 0.912 = 0.0329$ mm²; $d = \sqrt{(4 \times 0.0329 \div \pi)} =$ **0.205 mm**.
2. $T = \tan 30° \div (\pi d) = 0.577 \div (\pi \times 0.00020\ m) \approx$ **919 turns/m**.
3. $tpi = 3.5 \times \sqrt{20} = 3.5 \times 4.47 =$ **15.7 tpi** ≈ 15.7 × 39.37 ≈ **617 turns/m**. The coarser yarn has a larger diameter, so fewer turns per metre produce the same helix angle ($\tan\theta = \pi dT$): it is angle, not turns, that the multiplier holds constant.
4. θ increases with radius ($\tan\theta = \pi(2r)T$ for a fibre at radius r), so surface fibres are the most inclined and the most extended for a given yarn strain; they reach breaking extension first, shed their load onto the next layer, and failure propagates inward — the observed outside-in rupture of staple yarns.